

Release notes

26.5.20

v1.3.7

- 1: add command “sync time xx:xx yy:yy\n” in the protocol for syncing the time of lichess or chess.com with the robot.
- 2: add command “display txt *****\n” in the protocol for app to show text on robot’s screen.
- 3: improve the function of emoji displaying.
- 4: add double click to the clock button, a command “new game\n” will be sent out via BLE.
- 5: optimize the algorithm on rook detection.
- 6: add audio “ checkmate” for both side.

26.4.27

v1.3.6

- 1: improve algorithm of the illegal move detection.
- 2: fix the bug in en-passant detection.
- 3: fix the bug in castling.
- 4: modify the algorithm on illegal move, the robot will not continue if an illegal move is made by user.

26.4.23

v1.3.5

- 1: improve function of the castling record.
- 2: improve function of the en-passant record.
- 3: fix the bug that the robot can only detect illegal move one time.

26.4.17

v1.3.4

- 1: fix the conflict between firmware and APP on the color setting. The color can only be set by the APP when the BLE is connected, the auto color detection of the firmware is turned off.
- 2: optimize the promotion color detection algorithm.
- 3: fix the bug that the color is not set on the color detection.
- 4: change the background image of the time mode.
- 5: add protection to the measuring of connection between robot and board, fix the bug that the arm may reset due to jitters on the connection between robot and chessboard.
- 6: optimize the arm control algorithm, fix the bug that the arm may shake when it extends out.

2026.4.14

v1.3.3

- 1: add command “scan board\n” in the protocol. You can force the robot to scan the board by this command without triggering a robot move.
- 2: add protection on the board detection each time the board gets connected.
- 3: improve the color detection of the promotion queen.

2026.4.8

v1.3.2

- 1: add command “play audio **\n” in the protocol to let the robot play corresponding audio. There are only three audios supported now, “check”, “checkmate” and “error”
- 2: add command “get robot status\n” in the protocol.
- 3: add command “move AxB y s t\n” in the protocol.
- 4: fix the bug that the FEN is incorrect after a special move en-passant, castling, and promotion.
- 5: fix the bug that the FEN is incorrect on an under promotion.

2026.3.30

v1.3.1

- 1: add command “set wait minutes **\n” to change the time that robot entering the sleeping mode.
- 2: change the return of the “get battery\n”, the voltage of the battery will be ended with “+” if charging and “-” if not.
- 3: support upper case in the move command.

2026.3.19

v1.3.0

- 1: new model for chess pieces recognition, increase the accuracy. fix the bug that the pawn maybe recognized as knight or bishop in some scenario.
- 2: add audio “check mate” when the robot gets checked.
- 3: add promotion queen color detect, user can set the robot playing white by placing a white queen on the promotion area before the board is connected to the robot. When the robot gets connected, it will scan the board and if the white queen is detected in the promotion area, the robot will play white, the board will be flipped logically, which means the row 8 is by the user side. Otherwise the robot will play black by default.



- 4: fix the bug that changing difficulty may failed in some situation.
- 5: add command “set robot turn\n”, by this command, developer can make the robot scan the board just like pressing the clock button on the board.
- 6: fix some other bugs.

2025.11.26

v0.9.0

- 1: improve the piece recognition accuracy.
- 2: added upgrade function

3: add voice for check

4: fix some bugs

2025.11.26

v0.8.0

1: improve the piece recognition accuracy.

2: fix the bug that the left area of the chess board is only updated in the test mode, which cause the robot cannot place the captured chess piece correctly in a normal mode.

2025.11.14

v0.7.0

1: improve the piece recognition accuracy.

2025.11.13

v0.6.0

1: change the logic of updating the position information, the information is push to BLE host once it gets changed.

2025.11.4

v0.5.0

1: add information like check, illegal move, display on the screen.

2: new model for chess piece recognition.

3: add acknowledgment to the protocol to prevent the package lost.

2025.9.9

v0.4.0

1: new model for chess piece recognition, increase the recognition accuracy.

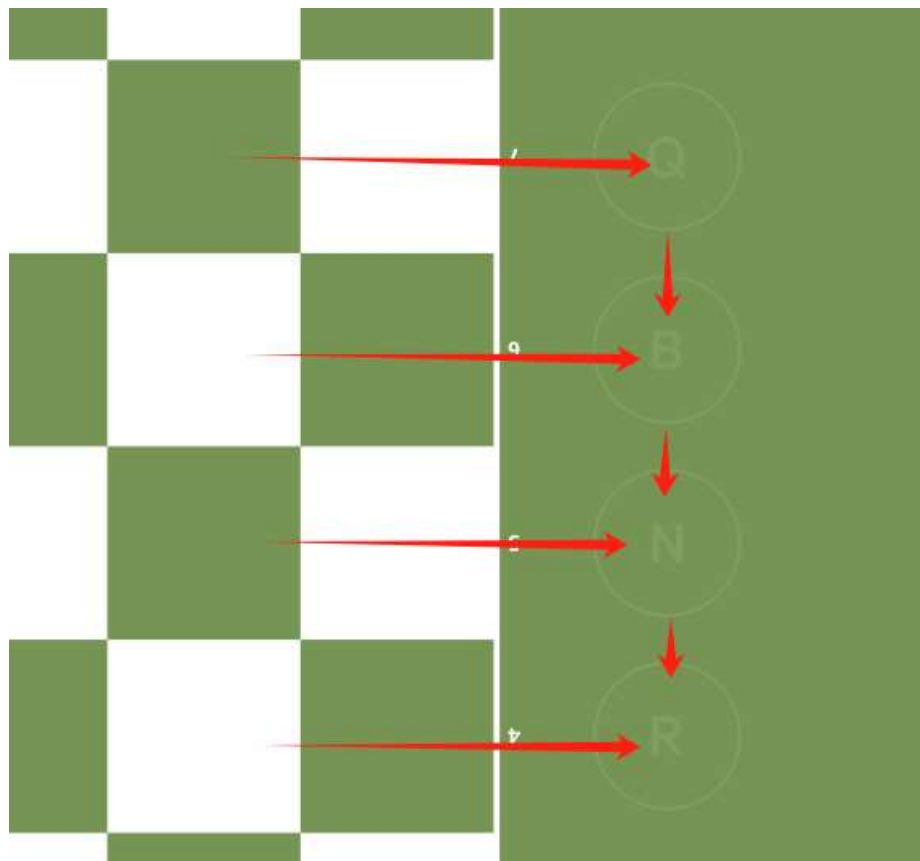
2: fix the bug that robot may drop a piece on the place where already has one placed.

2025.8.8

v0.3.0

1: fix the bug that queen and king cannot be move to h1 and a1.

2: add promotion to bishop, knight and rook. We will add marker for bishop, knight or rook on the chess board in the mass production. You can just put the piece on the corresponding place to make the robot work.



3: add audio for check mate.

4: disable the continuation on time out, and also add an audio as a warning.

Resetting the time or changing to unlimited mode is needed to get continue.